MILITARY SPECIFICATION

ROLL COTTON TAPE, RADIOPAQUE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rolled radiopaque cotton tape.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-276	Thread, Cotton.
DDD-T-86	Tape, Textile, Cotton, General Purpose (Unbleached, Bleached Or Dyed).
PPP-B-566	Box, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6510

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STANDARDS

Federal

FED-TEST-METHOD STD NO. 191	Textile Test Methods.
FED-TEST-METHOD STD NO. 601	Rubber, Sampling and Testing.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES PHARMACOPOEIAL CONVENTION, INC.

Pharmacopoeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Tape. Tape shall be Type I, class 2, 1/2 inch wide, in accordance with Fed. Spec. DDD-T-86.

3.1.1.1 Weight per linear yard. Weight per linear yard shall be 0.061 ounces minimum.

3.1.2 Thread. The thread shall be type IA-2, 3-ply, ticket No. 60, of Federal Specification V-T-276.

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3.1.3 Radiopaque filament. The radiopaque filament shall be made of vinyl plastic, containing 61 ± 3 percent of U.S.P. X-Ray grade Barium Sulfate (4.4.11). The filament shall be 0.022 - 0.033 inches in diameter. When tested as specified in 4.4.8 - the filament shall show no signs of bleeding or deterioration.

3.2 Construction. The tape and radiopaque filament shall have the following minimum dimensions when extended:

Tape - 3-1/8 inches
Filament - 5/8 inch

The tape and filament shall be assembled, folded, rolled and sewn into a roll as shown in Figure 1, using any of the constructions shown. No cut ends of tape shall be exposed. The filament shall be visible through the one layer of tape covering it. The number of stitches used to secure the folded edge of the tape to the tape roll body shall be at least 6. At least one end of the thread shall be knotted to prevent loosening of stitches. As an alternate to the knot, a "knotting stitch" shall be used at one end of the stitch line. Thickness of the finished tape roll shall be 1/4 inch ± 1/16 inch along any diameter. Overall length shall be not less than 1/2 inch, nor more than 5/8 inch.

3.3 Workmanship. The cotton - tape roll shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the components, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The unit of product for Major and Minor A defects shall be one roll; and for Minor B, one package.

Table I.

Type Defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor A	II	2.5
Minor B	S-3	2.5

4.2.2 For tests. The requirements and methods of testing shall be as listed in table II. The sampling level shall be S-1 with an acceptance number of zero. Minimum sample shall be three. Where possible, the same sample unit shall be used for the determination of two or more tests.

Table II. Tests

Characteristic	Test Requirement Paragraph	Test Method Paragraph
Tape		
Width	3.1.1	4.4.1
Length	3.2	4.4.2
Thread Count	3.1.1	4.4.3
Warp Breaking Strength	3.1.1	4.4.4
Weight Linear Yard	3.1.1.1	4.4.5
Radiopaque Filament		
Diameter	3.1.3	4.4.6
Length	3.2	4.4.7
Resistance to Sterilization	3.1.3	4.4.8
Barium Sulfate Content	3.1.3	4.4.11
Roll		
Length	3.2	4.4.9
Thickness	3.2	4.4.10

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4.3 Examination. The tape rolls shall be examined in accordance with the following classification of defects. Examination shall not be limited to the defects listed.

Table III.

Categories	Defects
Major A	
101	Cotton tape rolls not free from dirt or other foreign matter.
102	Cotton tape rolls not one continuous piece.
103	Both edges of tape not selvaged.
104	Cotton Tape roll not free of holes tears or cuts.
105*	Radiopaque filament missing from tape roll.
Minor A	
201	Cotton - Tape roll not evenly rolled.
202	Cotton - Tape roll not tightly rolled.
Minor B (Unit of product - package)	
251	Labeling or printing on package not legible and free from defacing marks or scratches.
252	Bag (containing 100 rolls) torn.
253	Count of rolls per bag not correct.

* For this defect, acceptance number shall be zero.

4.4 Tests.

4.4.1 Width. The width of the cotton tape shall be determined according to method 5020 of Federal Test Method Std. No. 191.

4.4.2 Length. Length of the cotton tape shall be determined according to method 5010 of Federal Test Method Std. No. 191, except paragraph 5.3.1 of the test method shall not apply. The cut pieces shall be placed flat, without tension on a smooth horizontal surface.

4.4.3 Thread count. The thread count (warp and fill) shall be determined in accordance with method 5050 of Federal Test Method Std. No. 191.

4.4.4 Warp breaking strength. The warp breaking strength of the tape shall be determined in accordance with method 4108 of Federal Test Method Std. No. 191.

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4.4.5 Weight per linear yard. Weight per linear yard shall be determined in accordance with method 5040 of Federal Test Method Std. No. 191, except as follows:

1. No width determinations need be made.

2. Weight shall be reported in ounces per linear yard.

4.4.6 Diameter of radiopaque filaments. Diameter of the radiopaque filaments shall be determined by a suitable strong optical measuring device.

4.4.7 Length of radiopaque filaments. Length of radiopaque filaments shall be determined in accordance with Method 2411 of Fed. Test Method Std. No. 601.

4.4.8 Resistance to sterilization of radiopaque filaments. Radiopaque filaments shall be sterilized by live steam at a temperature of 250° F. for one hour. After sterilization, the radiopaque filament shall meet the requirements of paragraph 3.1.3.

4.4.9 Length of tape roll. Length of tape rolls shall be determined using a ruler or metal tape graduated to read in 1/16 inch.

4.4.10 Thickness of tape roll. Thickness of tape rolls shall be determined by using a ruler or metal tape graduated to read in 1/16 inch.

4.4.11 Barium sulfate content, radiopaque filament. Ash 1 yard of the sample in a tared porcelain dish until white and weigh. Transfer to oven at 1500° F. and leave it there for 3/4 hr. Cool and place on hot plate under hood and add 1 cc of sulfuric acid. Place back in oven at 1500° F. for 15 minutes, cool to room temperature and weigh.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package containing 1000 rolls, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	12 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Immediate container. One hundred (100) rolls shall be packaged and sealed in a suitable bag.

5.1.3.2 Unit package. Ten (10) immediate containers (1000 rolls) shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening.

5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Fiberboard boxes shall conform to the special requirements specified in Table II of PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

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5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements specified in Table I of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Immediate container and Unit package. Immediate container and unit package shall be marked with item identification, federal stock number, name and address of manufacturers, contract or purchase order number, date of manufacture and the legend: NOT STERILIZED (or a similar statement). The unit package shall also bear the level of packaging, unit of issue and quantity.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. Date of manufacture shall be shown in lieu of the date packed.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Date of manufacture shall be shown in lieu of the date packed.

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6. NOTES

6.1 Intended use. The rolled radiopaque cotton tape covered by this specification is intended for use as an appendix sponge in blunt dissection.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the items which are normally procured to meet Federal Government requirements.

6.4 The roll cotton tape covered by this specification is listed in the Federal Supply Catalog under the following stock number:

Federal Stock Number	Item Identification
6510-114-0091	ROLL COTTON TAPE, RADIOPAQUE

6.5 Supersession data. This specification includes the requirements of Defense Medical Purchase Description No. 3, dated 4 May 1972.

Custodians:
Army - MD
Navy - MS
Air Force - 03

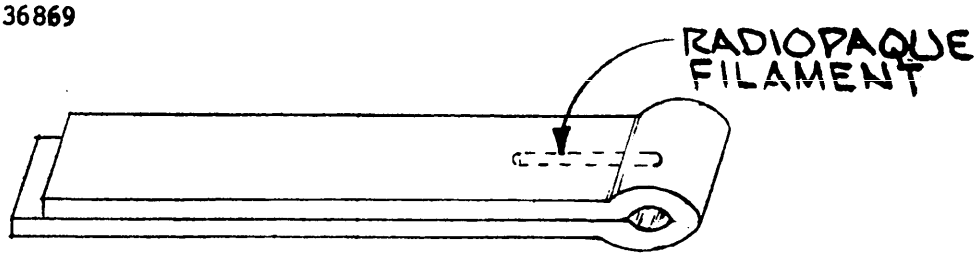
Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

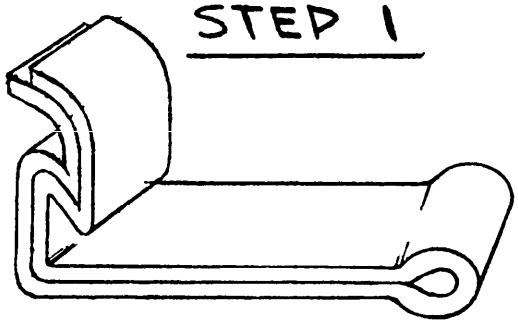
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS

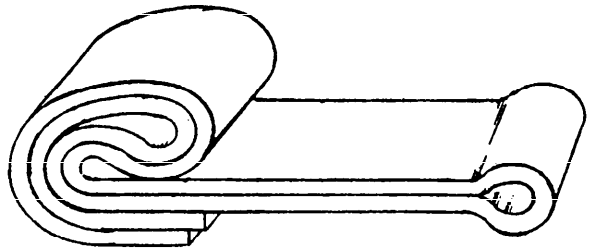
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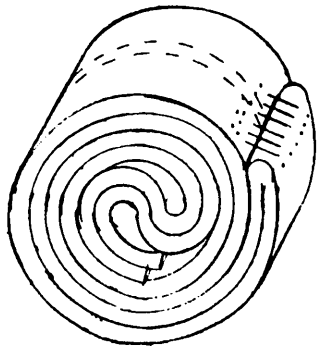
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STEP 3

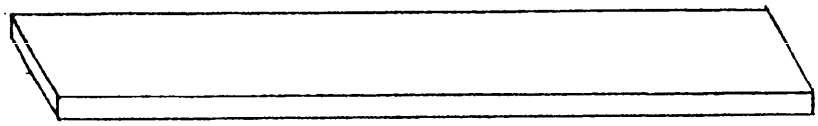


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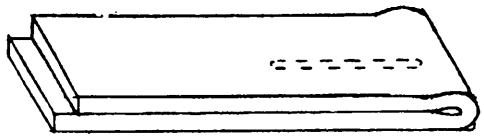
STEP 4

FIGURE 1

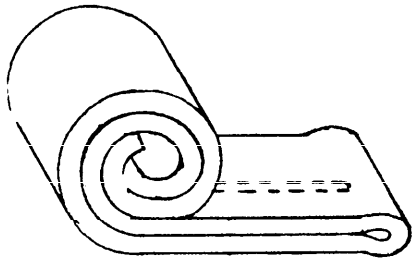
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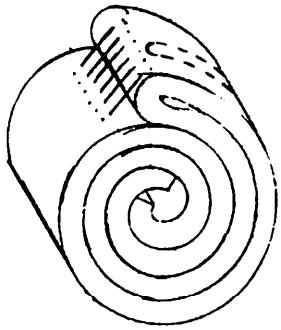
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STEP 2



STEP 3

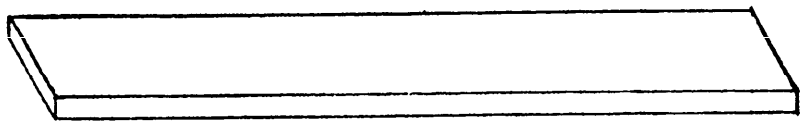


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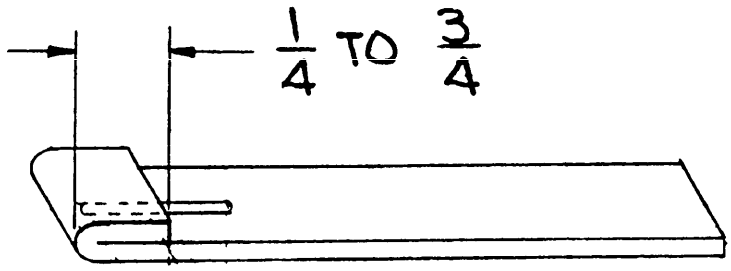
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FIGURE 1

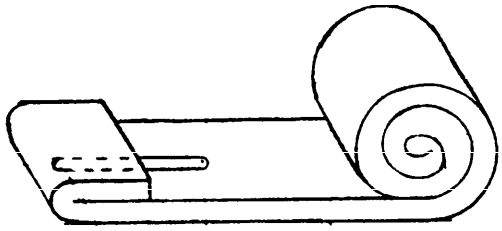
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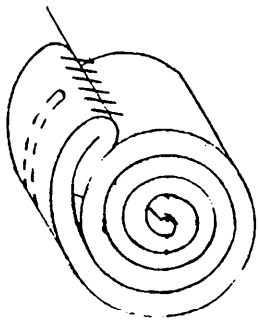
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STEP 3



ALTERNATE CONSTRUCTION #3

STEP 4

FIGURE 1

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OMB Approval
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MIL-R-36869

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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